



Photometric Determination of the Age and Distance of NGC 2281

A final report for the course NWI-NP031C, "Observational Astronomy"

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Chapter 1

Introduction

This report presents the observations, data reduction, and isochrone fitting used to determine the age and distance of the open cluster NGC 2281 (the “Broken Heart Cluster”), together with complementary measurements of the sky brightness in Nijmegen.

The Broken Heart Cluster is an open cluster found in the constellation of *Auriga*. It has a distance of 528^{+32}_{-28} pc and an age of 656 ± 28 Myr [40]. With an apparent magnitude of 5.4, a diameter of 25', and a high similarity to numerous open clusters strung across the Milky Way, the cluster lends itself well to ground-based observations [14]. In this project, a Colour-Magnitude Diagram (CMD) of NGC 2281 is created from measurements taken in the Sloan g' and r' filters, which is subsequently analysed using isochrone fitting to determine its age and distance.

Other than generating a CMD of NGC 2281, the sky brightness in Nijmegen is also measured. These secondary measurements are made to contribute to records of light pollution and weather conditions in Nijmegen which can be used by third parties to analyse how these variables change over time.

This report serves as a final product for the course NWI-NP031C, “Observational Astronomy”, at Radboud University Nijmegen, taught by Dr. S.S. Larsen.

Chapter 2

Observations

In this section, the procedure for observing NGC 2281 is outlined. Specifically, a clarification on the selection of NGC 2281 as the focus of our study is given, several important characteristics of the cluster are highlighted, and the conditions under which the observations were conducted are detailed.

2.1 On the Selection of NGC 2281

As a final product for the course NWI-NP031C at Radboud University Nijmegen, this report was required to focus on an independently selected cluster. To streamline the selection, the group established a set of requirements that each member used to identify candidate clusters. From these options, NGC 2281 was chosen collectively because it satisfied all established criteria and best matched the group's preference. The requirements are listed below.

1. The cluster should maintain an altitude of at least 45° between 19:00 and 24:00 during the February-March observing period.
2. The cluster should have an apparent magnitude of no greater than 6.0.
3. The cluster should have an apparent angular size between $10'$ and $30'$.
4. The cluster should have several stars for which magnitude measurements in the used filters are available in the URAT1 catalogue [45].

These criteria ensured that the selected cluster would be observable within the designated observation window while maintaining a sufficiently low airmass, thus reducing the effects of atmospheric disturbance and extinction (requirement 1), be adequately bright to be measured by the CCD camera with minimal contamination from external sources (requirement 2), have an apparent angular size compatible with the field of view of the camera such that contamination from non-member stars is minimised (requirement 3), and include stars with available catalogue magnitudes for calibration (requirement 4). Section 2.2 shows how NGC 2281 satisfied these requirements.

2.2 On the Characteristics of NGC 2281

According to Fisher [14], NGC 2281 is an open cluster in the constellation of Auriga located at R.A. 6 h 48 min 17.00 s, Dec. $+41^{\circ} 4' 42.0''$. It has an apparent magnitude of 5.4 and a diameter of 25' [14], thereby satisfying requirement 2 and requirement 3, respectively. The cluster has a distance of 528^{+32}_{-28} pc, and an age of 656 ± 28 Myr [40]. Furthermore, as shown in figure 2.1, the altitude of NGC 2281 stays above 45° during the planned observation window, with a minimum altitude during the date of observations of around 54° . Thus, the cluster also satisfies requirement 1.

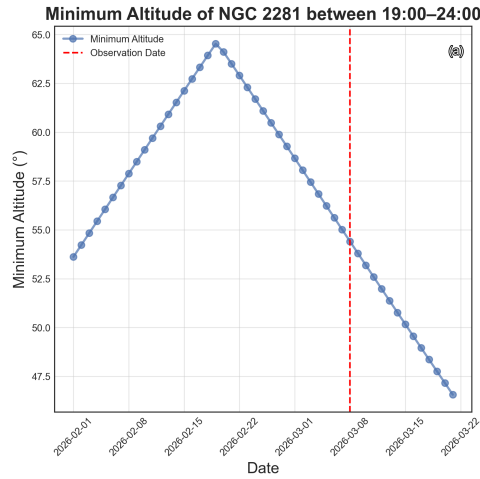


Figure 2.1 The minimum altitude in degrees reached by NGC 2281 during the 19:00–24:00 observation window in February and March. The red vertical dotted line on the eighth of March represents the observation date. The data was obtained using the astropy package [1]. The figure was generated using code from Leenaerts [26].

Finally, the cluster contains approximately 1100 stars [14], of which 35 have available magnitude measurements in the URAT1 catalogue [44], thereby satisfying requirement 4.

2.3 On the Conducted Observations

Observations were conducted on the eighth of March 2026, at the Radboud Universiteit faculty of science building located in Nijmegen, Netherlands ($51^{\circ}50'31''$ N, $5^{\circ}51'8''$ E). This section describes in more detail the process of making these observations.

2.3.1 Telescope

The observations were performed using a catadioptric 35.6 cm Schmidt-Cassegrain telescope with a focal distance of 3.56 m ($f/10$). Furthermore, a SBIG-ST-10XME CCD camera with a KAF-3200ME detector cooled to 5°C with a 2184×1472 (3.2 Mpx) detector and a pixel size of $6.8\ \mu\text{m}$ [10] was used. All images were acquired using 1×1 binning, corresponding to the native resolution of the CCD detector. Using this setup, a plate scale of 0.39 arcsec/px was achieved. Observations were conducted using the SDSS photometric Sloan g' and r' filters. The g' filter is characterized by a

bandpass spanning approximately 400 nm to 550 nm, corresponding to the blue-green region of the spectrum, while the r' filter is characterized by a bandpass spanning approximately 550 nm to 700 nm, covering the red region of the spectrum [16]. The filter responses of these filters are shown in figure 2.2.

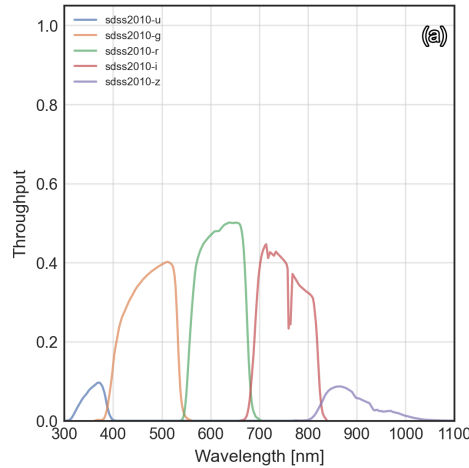


Figure 2.2 Response curves of the SDSS photometric filters as a function of wavelength, where each colour corresponds to a different filter. The data was obtained from Doi et al. [12]. The figure was generated using code from Leenaerts [26].

2.3.2 External Conditions

The observations were conducted on the eighth of March 2026 between 20:56 and 22:24. There were no visible clouds, with a reported coverage of 2 % according to Weer1.com [43]. Furthermore, during the observations, temperatures fell from 12 °C to 10 °C and humidity rose from 60 % to 68 %. Visibility was reported to be over 10 km [43].

As shown in figure 2.3, the angular separation between the Moon and NGC 2281 did not fall below 128°, and the lunar elevation remained strictly negative. Thus, no significant contribution from direct moonlight to the sky background was expected. The Moon was in its waning gibbous phase, with an illumination of 74 % [19].

Furthermore, figure 2.4 shows the evolution of the zenith distance and azimuth of NGC 2281 during the observations. The zenith distance remained below 25°, corresponding to an airmass of no more than 1.16 [22].

As will be shown in section 3.6, the sky brightness at the time of observations was around 18 mag/arcsec², which ranks under Class 8: Inner-City Sky on the Bortle Scale [5]. Finally, analysis of the images taken using SAOImageDS9 [37] gives a full width at half maximum (FWHM) of 5.7 px, or by multiplying by the pixel size as described in section 2.3.1, a FWHM of 2.22'', which corresponds to moderate seeing conditions.

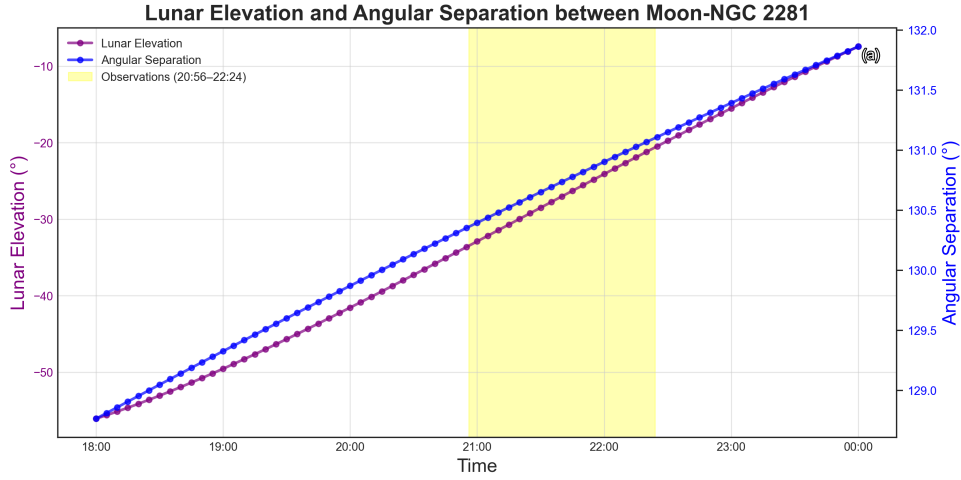


Figure 2.3 The angular separation between the Moon and NGC 2281 together with the lunar elevation on the eighth of March 2026 between 18:00 and 03:00. The colored region shows the time frame in which images were taken. Data was obtained using the astropy package [1]. The figure was generated using code from Leenaerts [26].

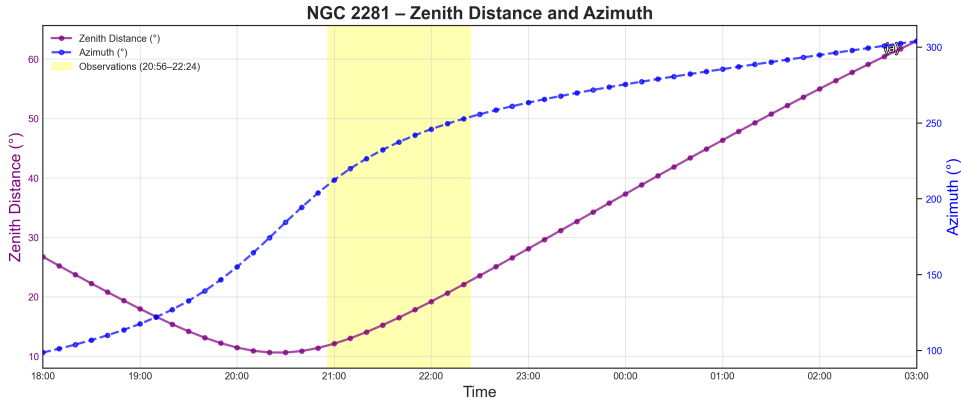


Figure 2.4 The zenith distance of NGC 2281 along with its azimuth on the eighth of March 2026 between 18:00-03:00. The colored region shows the time frame in which images were taken. Data was obtained using the astropy package [1]. The figure was generated using code from Leenaerts [26].

2.3.3 Measurements

A total of 15 images of NGC 2281 were obtained per filter. Furthermore, 10 flat-fields were taken for each filter from the inside of the dome. Taking separate dark frames was deemed unnecessary, as the provided option in which the camera obtained and applied these automatically was used.

An exposure time of 30 s was chosen for the images, as it achieved a sufficiently high signal-to-noise ratio for photometric analysis while avoiding detector saturation. For taking the flats, however, an exposure time 3 s was used to prevent over-saturation of the detector. This resulted in a total integration time of 450 s for the images of NGC 2281, and an integration time of 30 s for the flat fields. Images were exported in the FITS file format.

Data Analysis

In this section, the process of analysing the data obtained according to section 2 is described. All measurement data was stored in the FITS format, which was subsequently processed using a Python Jupyter Notebook. All Python code used in this section can be found in the GitHub repository “Observational Astronomy” from Leenaerts [26].

3.1 Flat-Field Corrections

Firstly, it was necessary to process the flat-fields to map sensitivity variations in the CCD camera. As all flat-fields were assumed to have been taken from a uniformly illuminated surface, deviations of pixel values from the mean are a measure of relative sensitivity. To eliminate noise, multiple independent flat-fields were combined.

This was achieved by first converting each pixel value from ADU to counts by adding the PEDESTAL value stored in the header of each FITS file, and subsequently normalising each image to a mean pixel value of one. For this, algorithms from the Astropy [1] and CCDProc [11] packages were used. This normalisation enabled the combination of flat-fields by averaging counts pixel-wise across images, creating a so-called “master-flat” in which each pixel value corresponds to relative sensitivity. By dividing measurements of NGC 2281 pixel-wise by these master-flats, instrumental sensitivity deviations were mitigated.

The master-flats are shown in figure 3.1 and an example of applying this correction to a measurement of NGC 2281 is shown in figure 3.2.

3.2 Combining Measurements

After having corrected the raw measurements for sensitivity variations, they were aligned to an arbitrarily chosen measurement using the astroalign package [4]. This procedure ensured proper alignment of stars between the multiple measurements taken using each filter.

Subsequent combination of measurements was performed in the same manner as for the flat-fields: each pixel value was converted from ADU to counts by adding the PEDESTAL, after which counts were averaged pixel-wise. This procedure again mitigated noise and increased equivalent exposure time. The result for each filter is shown in figure 3.3.

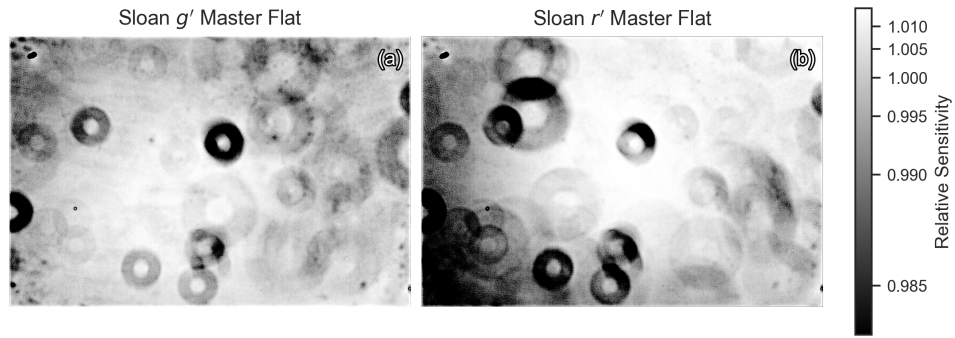


Figure 3.1 The combined master-flats for both the g' and r' filters. Brightness is used to indicate the sensitivity deviation from the mean for both the g' and r' filters.

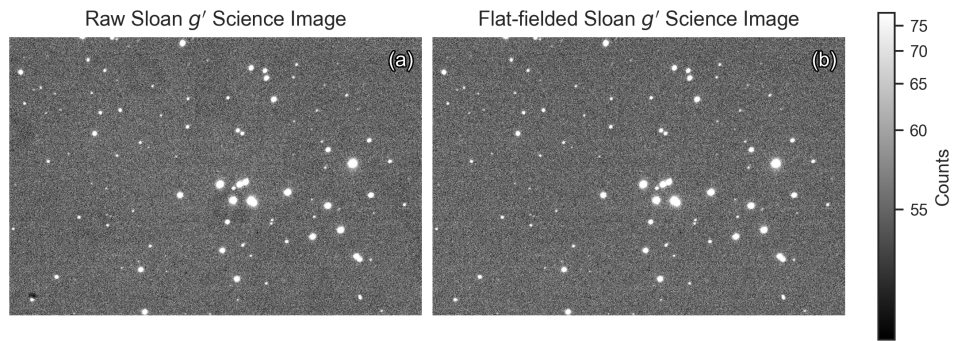


Figure 3.2 An example for how a raw observation of NGC 2281 is corrected with a master-flat to produce a sensitivity deviation corrected image. Note how several "donut"-shaped darker regions in the original are corrected in the flat-fielded image.

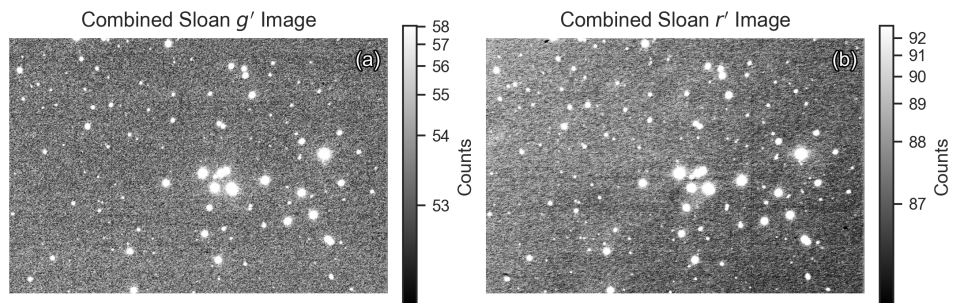


Figure 3.3 The combined measurements of NGC2281 in both the g' and r' filters.

3.3 Extracting Background and Sources

After the raw data had been reduced to two images, the final reduction step was the extraction and subtraction of a background model. This model was extracted using the `Background` class from the SEP (Source Extraction and Photometry) package [3]. A background box size of 100 px and a filter width and height of 5 were used. The result is shown in figure 3.4.

At this point, all data had been reduced, allowing preparation for photometry. The first step was to extract sources (stars) from both images. For this, the `extract` method from the SEP package was used. The method gave the best results with a detection threshold of 2σ and a minimum area of

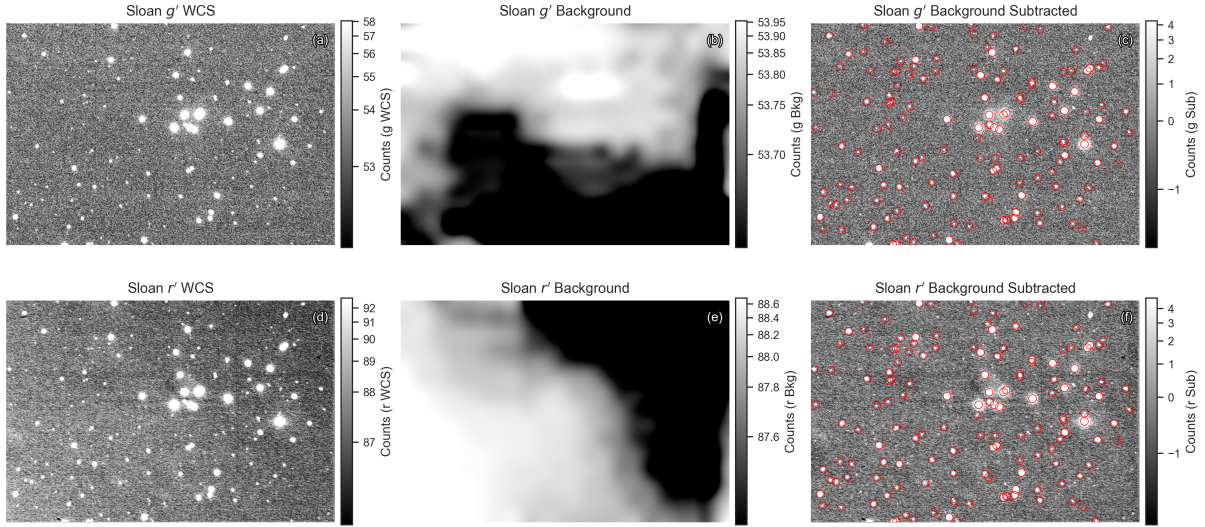


Figure 3.4 The input images, extracted background model, and background-subtracted output image for both filters. For extracting the background model, a background box size of 100 px and a filter width and height of 5 were used. Red circles in the background-subtracted images indicate extracted sources.

5 px. This procedure resulted in 154 identified sources in both filters. The detected sources in both images are shown in figure 3.4.

After source extraction, uncalibrated fluxes and their associated errors were computed for both filters using the `sum_circle` method from the SEP package. An aperture of 11.4 px, exactly twice the FWH that was found in section 2.3.2, and a gain of 1.36, retrieved from the FITS header, were used. These fluxes and errors were subsequently converted to instrumental magnitudes, which were used to construct an initial uncalibrated Colour-Magnitude Diagram, shown in figure 3.5.

3.4 WCS and Zero Point

Final calibration of both images was achieved by first using Astrometry.net [25] to add World Coordinate System (WCS) information to the headers of both images. This information was then used to convert pixel coordinates of each extracted source to equatorial coordinates (RA, Dec) using the Astropy package.

Once a list of sources, their equatorial coordinates, and their fluxes with associated errors had been obtained, sources were matched to entries in the URAT1 catalogue [45]. This catalogue contains apparent-magnitude photometry of stars in the northern sky and can therefore be used for zero-point estimation. The overlap between detected sources and URAT1 entries is shown in figure 3.6.

Matching was done based on the smallest distance between sources and URAT1 entries for which both g' and r' magnitudes were available, excluding matches in which the distance exceeded $1''$. This procedure resulted in 18 found matches.

From these matches, photometric zero points were computed as the weighted average of the difference between URAT1 and extracted magnitudes according to their errors, rejecting values that

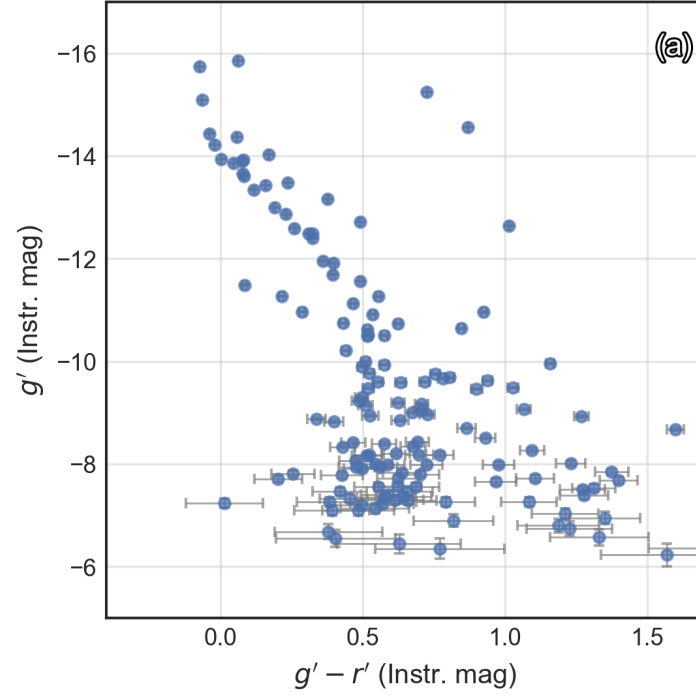


Figure 3.5 Uncalibrated Sloan g' and r' Colour Magnitude Diagram for NGC 2281 along with the errors for each source.

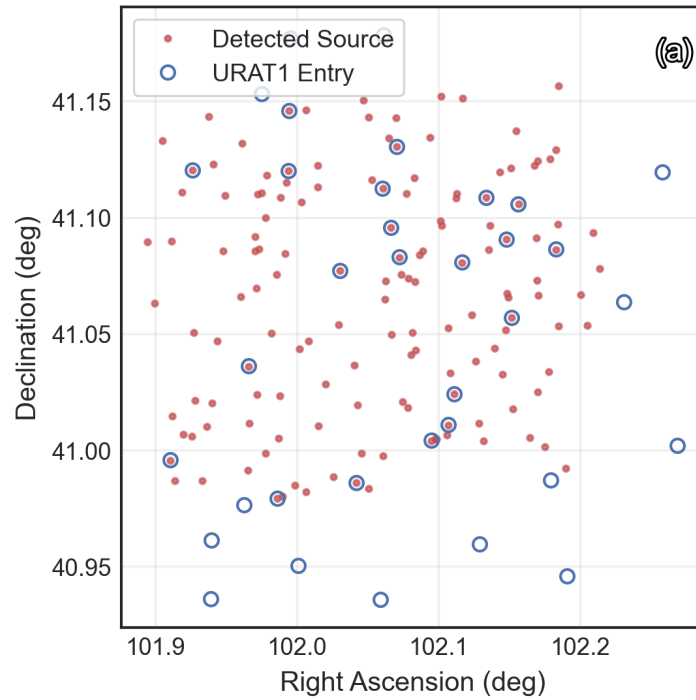


Figure 3.6 Detected sources and URAT1 [45] entries plotted according to their equatorial coordinates.

deviated more than 2σ from the mean. This resulted in zero-points for the Sloan g' and r' filters of

24.612 ± 0.012 mag and 24.599 ± 0.008 mag respectively. A graphical representation of this procedure is shown in figure 3.7, which shows a high level of agreement between different zero-point estimations, with a single outlier to the bottom right of both filters. This outlier is the main motivation behind the usage of the 2σ rejection.

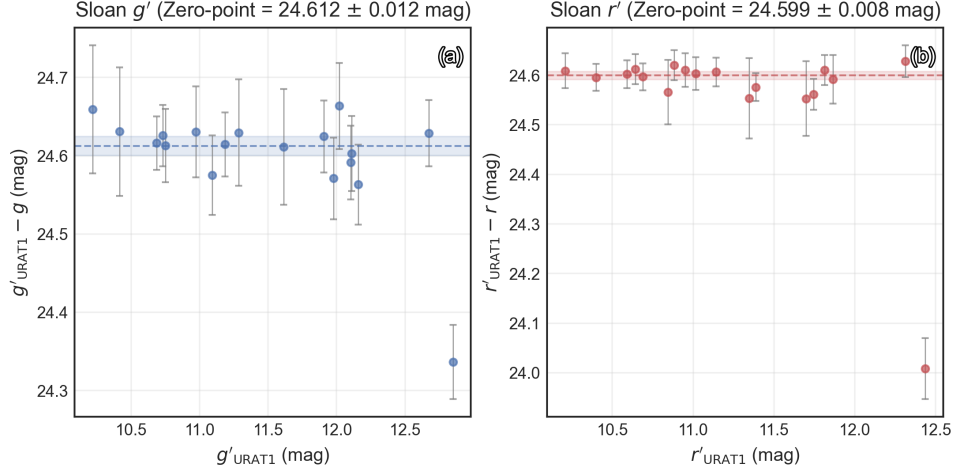


Figure 3.7 The difference between the URAT1 and extracted magnitude for all sources and their respective errors in both the Sloan g' and r' filters. The vertical dotted line corresponds to the zero-point for that filter computed from the weighted average of the magnitude difference according to the error of the matches after having rejected values that deviated more than 2σ from the mean. The shaded region shows the error on the estimated zero-point.

This allowed the generation of a calibrated Colour-Magnitude Diagram as shown in figure 3.8.

Finally, using empirical color transformations from Jordi, Grebel and Ammon [23], the Sloan g' and r' color-magnitude diagram was converted into a standardised Johnson B and V CMD. These transformations, however, introduce additional uncertainties as they are but approximations, so this diagram is presented primarily for the purpose of comparison with existing studies.

3.5 Photometry

Once a CMD for NGC 2281 had been constructed, its distance and age could be estimated through photometric analysis. This was achieved through isochrone fitting using models from the MESA Isochrones & Stellar Tracks (MIST) database [13]. Synthetic isochrones computed for the SDSS photometric system were adopted, assuming a rotation rate of $v/v_{\text{crit}} = 0.4$ and a metallicity of $[\text{Fe}/\text{H}] = 0$.

Several assumptions underlie this procedure. Firstly, a rotation rate of $v/v_{\text{crit}} = 0.4$ was adopted, implying that stars rotate at 40 % of their critical (break-up) velocity. This value corresponds to moderate stellar rotation and is the default value for MIST models [13]. Furthermore, a metallicity of $[\text{Fe}/\text{H}] = 0$ was assumed, corresponding to a solar iron-to-hydrogen abundance ratio. This choice is supported by observational studies from Fritzewski, D. J. et al. [15] indicating that NGC 2281 has a near-solar metallicity.

Fitting isochrones to the extracted data consisted of three independent steps. Firstly, zero-point

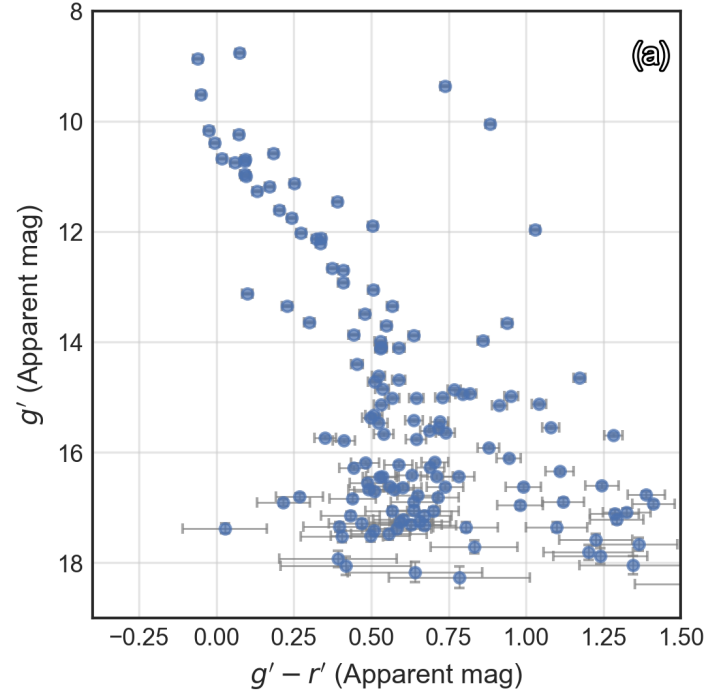


Figure 3.8 Calibrated Sloan g' and r' Colour Magnitude Diagram for NGC 2281 along with the errors for each source.

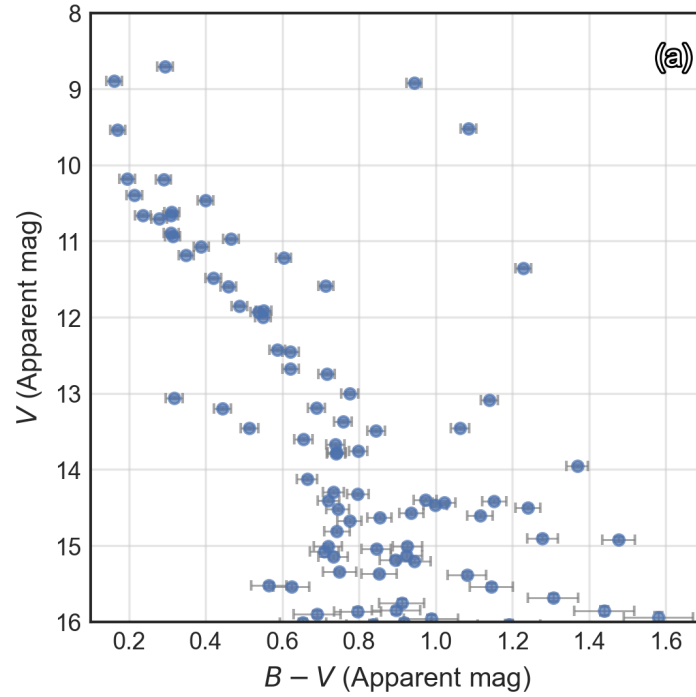


Figure 3.9 Calibrated Johnson B and V Colour Magnitude Diagram for NGC 2281 along with the errors for each source. Transformations between figure 3.8 were made using data from Jordi, Grebel and Ammon [23].

estimation using the URAT1 catalogue did not correct for interstellar extinction caused by scattering and absorption by dust and gas, which led to reddening and dimming of apparent magnitudes.

According to Terrell, Gross and Cooney [40], the reddening, $E(B - V)$, for NGC 2281 is around 0.110 ± 0.013 mag. Converting this using bandpass conversion data from Schlafly and Finkbeiner [36] for the Sloan g' and r' filters, extinctions of $A_g = 0.363 \pm 0.043$ mag and $A_r = 0.251 \pm 0.030$ mag were computed for the g' and r' filters respectively. These values were applied as a constant offset to the calibrated apparent magnitudes, effectively correcting them to what would be observed in the absence of interstellar dust.

Secondly, the extinction-corrected apparent magnitudes were transformed into absolute magnitudes by applying a vertical shift in the CMD corresponding to the distance modulus. This adjustment aligns the observed main sequence with the synthetic isochrone, allowing a direct determination of the cluster's distance.

Finally, the cluster age was estimated by selecting the isochrone that provides the closest match to the observed CMD. Special attention was given to the Main Sequence Turnoff Point (MSTO), as its position is highly sensitive to age and represents the primary feature distinguishing different isochrones.

The result of the above procedure is shown in figure 3.10. Note that extinction correction introduces a significant increase in error in the $g - r$ values for sources.

Errors on the derived quantities were conservatively estimated visually by the generated CMD. Firstly, as main-sequence magnitudes remain relatively unaffected by age, errors on distance modulus were derived from the maximum vertical shift for which this section of the isochrones agreed with the CMD. Adopting an uncertainty of $\Delta(m - M) = \pm 0.5$, the associated uncertainty in distance was computed to be 110 pc.

Conversely, as MSTO is affected mostly by cluster age, errors on this parameter were estimated by finding extreme values for which isochrones agreed with the CMD. Note that this resulted in a conservative estimate of the uncertainty due to the resolution of isochrones available in the dataset rendering a tight upper and lower bound fit unachievable. This procedure yielded a final distance of 479 ± 110 pc and age of 589^{+140}_{-120} Myr.

3.6 Sky Brightness

SAOImageDS9 [37] enabled sky-brightness estimation by providing functionality for computing average counts per pixel in starless regions of the combined images of NGC 2281. Previously determined zero points, as described in section 3.4, were then used to transform these measurements into apparent magnitudes, providing an estimate of the sky brightness above Nijmegen on the observation date.

For an accurate sky-brightness estimate, the above procedure was conducted on the combined NGC 2281 images for both filters, which had undergone pixel-wise averaging and flat-field correction but not background removal. This was deemed sufficient to reduce noise for accurate measurements. Multiple regions were analysed to estimate the sky brightness above Nijmegen on the observation date. The results are shown in table 3.1.

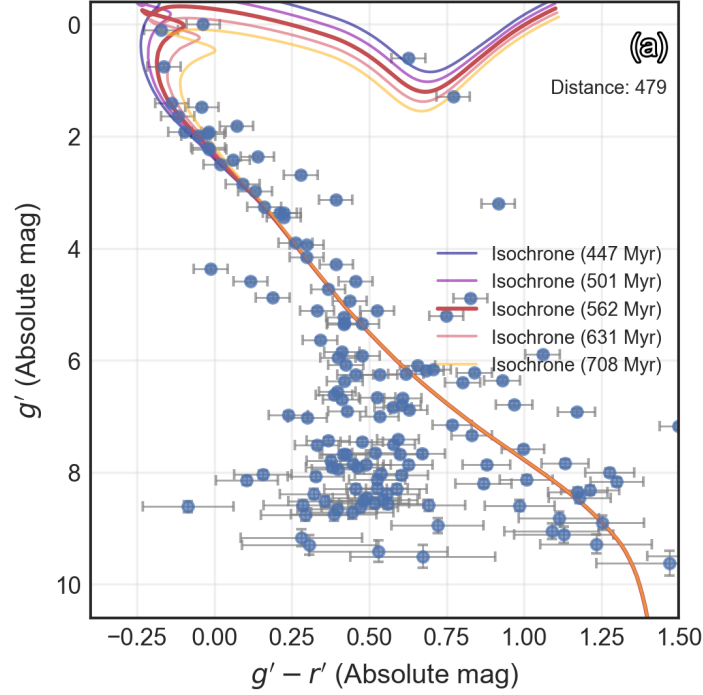


Figure 3.10 Extinction-corrected CMD along with its errors of NGC 2281, shifted by the distance modulus, with several overlaid synthetic isochrones. The best-fitting isochrone is highlighted in red, corresponding to a distance of 479 pc and an age of 589 Myr

From these measurements, the average counts per pixel in the Sloan g' and r' filters were determined to be 53.46 ± 0.06 counts/pixel and 83.71 ± 0.43 counts/pixel, respectively. To convert these values to apparent magnitudes, counts per pixel were first transformed into counts per square arcsecond using the known pixel scale described in section 2.3.1. The corresponding fluxes were then converted to magnitudes by applying the photometric zero points shown in figure 3.7.

A limitation to this procedure arises from the finite aperture used to measure stellar fluxes. As a fraction of stellar blurred light falls outside this aperture, the measured fluxes are underestimated and the subsequently derived zero-points overcompensate for this flux loss. In contrast, the sky background, being approximately uniform, is not subject to the same aperture-dependent flux losses as any adjacent “patch” of sky will blur light into the measured area, effectively compensating for this loss. Consequently, applying these zero-point corrections induces a systematic bias, resulting in

Filter	Region 1	Region 2	Region 3	Region 4
Sloan g' counts/pixel	53.377	53.521	53.596	53.341
Sloan r' counts/pixel	83.893	84.840	83.211	82.879
Sloan g' Average $\pm$ SEM	53.46 ± 0.06	—	—	—
Sloan r' Average $\pm$ SEM	83.71 ± 0.43	—	—	—

Table 3.1 Average counts per pixel measured in starless regions of NGC 2281 across combined master images for both used filters. The last rows show the mean values and standard error of the mean (SEM) for each filter.

a slight overestimation of the sky surface brightness.

For an aperture radius equal to twice the FWHM, approximately $f = 90 \pm 2\%$ of a star's total flux is captured [20], necessitating a finite-aperture correction factor for this effect. The corresponding magnitude calculation can be expressed as:

$$m = -2.5 \log_{10} \left(f \times \frac{\text{counts}}{\text{pixel scale}^2} \right) + ZP.$$

Applying this procedure, keeping in mind the propagating error on the computed photometric zero-points and correction factor, yields an average sky brightness of $18.36 \pm 0.03 \text{ mag/arcsec}^2$ in the g' filter and $17.86 \pm 0.03 \text{ mag/arcsec}^2$ in the r' filter. Analysis of error propagation reveals the uncertainty is dominated primarily by the finite-aperture correction factor. Using the Bortle scale for sky brightness [5], these magnitudes correspond to an inner-city sky brightness.

Conclusion

In this report the observations, data reduction, and isochrone fitting used to determine the age and distance of the open cluster NGC 2281 (the “Broken Heart Cluster”), together with complementary measurements of the sky brightness in Nijmegen, were presented.

Using isochrone fitting, an estimated distance of 479 ± 110 pc and age of 589^{+140}_{-120} Myr were found. Furthermore, a sky brightness of 18.36 ± 0.03 mag/arcsec² in the g' filter and 17.86 ± 0.03 mag/arcsec² in the r' filter were determined.

Primary sources of error were identified to be caused by uncertainty in the extinction correction, finite-aperture corrections, and error in zero-points.

Chapter 5

Discussion

This report succeeds in generating a Colour-Magnitude Diagram (CMD) for the open star cluster NGC 2281 from which its age and distance are estimated using isochrone fitting. As shown in section 3.5, this results in a distance of 479 ± 110 pc and age of 589^{+140}_{-120} Myr.

On the comparison with previous research The acquired values for age and distance align well within their respective errors with previous research by Terrell, Gross and Cooney [40] which found an distance of 528^{+32}_{-28} pc and an age of 656 ± 28 Myr.

Furthermore, a comparison between the generated CMD and external sources, as illustrated in figure 5.1, indicates a satisfactory level of agreement in their characteristic features. In particular, both the main sequence and its turnoff point exhibit clear similarities across the two diagrams.

However, discrepancies are evident at higher magnitudes. In the generated CMD, the main sequence appears more fragmented and exhibits increased scatter compared to that reported by Terrell, Gross and Cooney [40]. This divergence may be attributed to the absence of membership filtering of extracted sources.

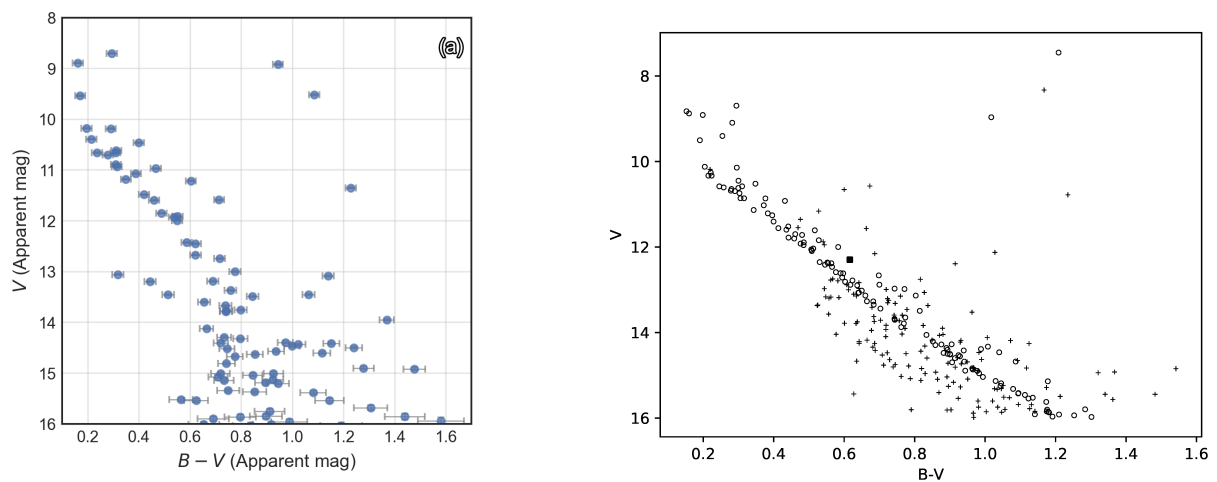


Figure 5.1 Comparison of the generated Johnson B-V CMD (left) with the Johnson $B - V$ CMD (right) obtained using data from Terrell, Gross and Cooney [40].

On the distance and age measurements The dominant source of uncertainty in the constructed color-magnitude diagram (CMD) was attributed to interstellar extinction, as the associated uncertainty in the extinction was estimated to be no less than 0.03 mag. Future work could mitigate this source of error by employing more sophisticated and precise methods for determining interstellar extinction, thereby improving the reliability of the derived parameters.

Photometric zero-points were estimated with errors of no more than 0.012 mag, which were insignificant compared to the already established errors on source fluxes and interstellar extinction. It is therefore not expected that any significant non-systematic error was introduced by zero-point corrections.

It is important to note that the quoted uncertainties in the extinction-corrected apparent magnitudes of the identified sources likely represent optimistic estimates, as these uncertainties account only for random statistical errors and do not include potential systematic effects. Sources of systematic uncertainty may include calibration errors, residual background variations, and imperfections in the photometric zero-point determination.

While these systematic deviations are expected to be captured by the conservativity of the errors on the cluster's age and distance, no such alteration to the errors on the sky brightness measurements have been made. As such, their true errors could be larger than reported.

There are, however, several caveats to the analysis presented in this report. The primary limitation concerns the reproducibility of the isochrone fitting, which was performed visually. Although χ^2 fitting was considered and attempted, all implementations of this method resulted in visibly poorer fits than those obtained manually. The underlying cause of this discrepancy was not investigated further, as visual fitting was deemed sufficient for the purposes of this study. Nevertheless, future work should focus on developing and validating robust algorithmic fitting procedures.

Additionally, cluster membership is not determined, and therefore not accounted for. This implies that stars external to NGC 2281 could be cluttering the generated CMD, and therefore could affect the subsequent analysis. However, since the field of view of the images closely matches the angular size of NGC 2281, any contamination from field stars is expected to be minimal and unlikely to significantly affect the results. The generated CMD as shown in figure 3.8 supports these expectations as a clear main sequence of stars is visible. However, cluttering near the bottom of the CMD is visible from sources with high errors that deviate from the main-sequence, which could be from non-member sources.

Furthermore, according to Terrell, Gross and Cooney [40] NGC 2281 contains at least several (eclipsing) binary stars. These binaries would result in exceptionally low magnitudes and therefore affect the generated CMD. However, it is expected that the number of these abnormal systems is low enough as to not substantially affect results. This is supported by the generated CMD, as no extreme low-magnitude outliers are visible.

Additionally, although interstellar extinction has been corrected for, a significant variation in extinction coefficients was observed between different papers. Any variation from the used values would affect the derived age and distance of NGC 2281. While this report employs modern, well-established extinction data, more accurate measurements could thus further improve these results.

Finally, all measurements were taken consecutively on the same evening, with a total integration time of 450 s. Observations on additional nights could help reduce instrumental and atmospheric

errors, increase signal-to-noise ratios for the combined images, and allow the detection of fainter cluster members through increased total integration time.

On the sky brightness measurements Although analysis of the sky brightness measurements with the Bortle scale placed them in the expected category of inner-city sky brightness, several sources of error were identified. Firstly, although finite-aperture effects were corrected for, uncertainty in the exact factor led to significantly increased errors in the final obtained values.

In addition, while general light pollution is an intended component of the measured sky brightness, stray light from nearby artificial sources such as streetlights, building illumination, or reflective surfaces may have locally increased background counts. This could have led to slightly brighter measured fluxes and therefore lower apparent magnitudes in affected regions. In particular, the nearby greenhouses could have severely contributed to this stray light.

Furthermore, the measurements were conducted over a period of approximately two hours prior to midnight. During this time, residual twilight or scattered sunlight may have contributed to the measured sky brightness. Averaging measurements over the entire night, or preferably across multiple nights, would help mitigate these effects and provide a more robust estimate of the typical sky brightness.

Finally, although the CCD camera used possesses a wide linear dynamic range, slight non-linearities at low flux levels cannot be entirely ruled out. Such deviations would not be corrected by zero-point calibration made from relatively bright stellar sources as this method assumes a strictly linear response, and could therefore introduce deviations in the measured sky brightness. Based on the specifications of the ST-10XME and its KAF-3200ME detector however, these effects are expected to be minimal.

Further research could reduce these sources of error by minimizing the bias in zero-points caused by the finite aperture in a more sophisticated way. This can be achieved by computing and applying an aperture correction factor, determined from the fluxes of stellar sources measured with multiple aperture radii. This correction would account better for the fraction of stellar light falling outside the used aperture and would allow better determination of zero-points and, in turn, of sky brightness.

Furthermore, a more robust method to correct for atmospheric extinction involves obtaining measurements of standard stars at multiple zenith angles, corresponding to different airmasses. The extinction coefficient can then be determined by performing a linear fit of the measured instrumental magnitudes against airmass, using reference magnitudes from the URAT1 catalogue. The resulting extinction coefficient, given by the slope of the fit, can subsequently be applied to correct sky brightness measurements for atmospheric effects, providing a more accurate determination of the true sky brightness above the atmosphere.

Chapter 6

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